KITABU QUEST

P6

SCIENCE AND ELEMENTARY TECHNOLOGY

Chameleon

Cover scene

learners doing a simple science and technology investigation with a chameleon mascot

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE 6

RWA REB-CBC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST P6 Science and Elementary Technology

Rwanda REB-CBC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps P6 learners study Science and Elementary Technology one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Observe, ask questions, design safely, record evidence, and explain findings.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

As you finish each topic, return to the success criteria and correct one answer before moving on.

Mechanics tools: Lesson Opener

Learning focus: Mechanics tools

Country link: a safe Rwandan observation, model, data table, or investigation for Mechanics tools.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- use mechanics tools correctly, take care of them, and act safely and responsibly when working with tools
- identify common mechanics tools
- explain the use of mechanics tools
- state the dangers of using tools in the wrong way
- use mechanics tools safely and correctly
- clean and store tools properly after use

Inquiry questions:
- How can mechanics tools help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Mechanics tools: Learn

Mechanics tools is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Mechanics tools
- Mechanics
- tools
- concept
- observation
- evidence
- safety
- conclusion

Mechanics tools: Worked Example

Investigation model for Mechanics tools: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Mechanics tools: Vocabulary And Study Skill

Use these words and study moves before you practise Mechanics tools. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Mechanics tools
- Mechanics
- tools
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Mechanics tools without a clear example, method, or evidence.

Mechanics tools: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Mechanics tools.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Mechanics tools: Practice And Correction

Practice for Mechanics tools:

Main outcome: use mechanics tools correctly, take care of them, and act safely and responsibly when working with tools.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Mechanics tools: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Mechanics tools.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Mechanics tools: Outcome Check

Outcome check for Mechanics tools: Use mechanics tools correctly, take care of them, and act safely and responsibly when working with tools.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Mechanics tools: Outcome Check

Outcome check for Mechanics tools: Identify common mechanics tools.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Simple machines: Lesson Opener

Learning focus: Simple machines

Country link: a safe Rwandan observation, model, data table, or investigation for Simple machines.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- identify, use, and care for simple machines in a safe and responsible way, and understand how they help make work easier
- define what simple machines are
- identify different types of simple machines
- distinguish between simple machines and other tools
- explain how each type of simple machine works
- group simple machines based on their classes

Inquiry questions:
- How can simple machines help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Simple machines: Learn

Simple machines shows how tools make work easier.

Core idea:
- A simple machine changes the size or direction of a force.
- Levers, pulleys, wheels and axles, inclined planes, wedges, and screws are simple machines.
- A machine does not remove work; it helps you use force more conveniently.
- Compare effort force, load, and distance moved.

Key words:
- Simple machines
- Simple
- machines
- concept
- observation
- evidence
- safety
- conclusion

Simple machines: Worked Example

Investigation model for Simple machines: Use a ruler as a lever.

1. Place a small object as the load.
2. Put a pencil under the ruler as the pivot.
3. Press the other end gently.
4. Move the pivot and compare the effort needed.

Conclusion: the position of the pivot changes how a lever helps lift a load.

Simple machines: Vocabulary And Study Skill

Use these words and study moves before you practise Simple machines. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Simple machines
- Simple
- machines
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Simple machines without a clear example, method, or evidence.

Simple machines: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Simple machines.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Simple machines: Practice And Correction

Practice for Simple machines:

Main outcome: identify, use, and care for simple machines in a safe and responsible way, and understand how they help make work easier.

1. Name three simple machines.
2. Identify the load, effort, and pivot in a lever.
3. Explain how changing the pivot changes the effort.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Simple machines: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Simple machines.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Simple machines: Outcome Check

Outcome check for Simple machines: Identify, use, and care for simple machines in a safe and responsible way, and understand how they help make work easier.

1. Name one simple machine.
2. Identify load, effort, and pivot or contact point.
3. Explain how it makes work easier.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Simple machines: Outcome Check

Outcome check for Simple machines: Define what simple machines are.

1. Name one simple machine.
2. Identify load, effort, and pivot or contact point.
3. Explain how it makes work easier.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Air pollution: Lesson Opener

Learning focus: Air pollution

Country link: a safe Rwandan observation, model, data table, or investigation for Air pollution.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- identify causes and dangers of air pollution, take action to reduce it, and show care for clean and healthy air in the environment
- identify common air pollutants in the environment
- explain the causes and dangers of air pollution
- identify signs of air pollution
- practice ways to reduce air pollution in daily life

Inquiry questions:
- How can air pollution help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Air pollution: Learn

Air pollution is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Air pollution
- pollution
- concept
- observation
- evidence
- safety
- conclusion

Air pollution: Worked Example

Investigation model for Air pollution: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Air pollution: Vocabulary And Study Skill

Use these words and study moves before you practise Air pollution. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Air pollution
- pollution
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Air pollution without a clear example, method, or evidence.

Air pollution: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Air pollution.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Air pollution: Practice And Correction

Practice for Air pollution:

Main outcome: identify causes and dangers of air pollution, take action to reduce it, and show care for clean and healthy air in the environment.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Air pollution: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Air pollution.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Air pollution: Outcome Check

Outcome check for Air pollution: Identify causes and dangers of air pollution, take action to reduce it, and show care for clean and healthy air in the environment.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Air pollution: Outcome Check

Outcome check for Air pollution: Identify common air pollutants in the environment.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Cows and goats management: Lesson Opener

Learning focus: Cows and goats management

Country link: a safe Rwandan observation, model, data table, or investigation for Cows and goats management.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- explain how to care for cows and goats properly and show good habits in keeping animals that help families and communities
- name different types of cows and goats and their uses
- state the characteristics of good cowshed or goat shelter
- explain how to protect cows and goats from common diseases
- choose healthy cows and goats for keeping or breeding
- use simple and correct words when talking about cows and goats

Inquiry questions:
- How can cows and goats management help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Cows and goats management: Learn

Cows and goats management is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Cows and goats management
- Cows
- goats
- management
- concept
- observation
- evidence
- safety

Cows and goats management: Worked Example

Investigation model for Cows and goats management: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Cows and goats management: Vocabulary And Study Skill

Use these words and study moves before you practise Cows and goats management. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Cows and goats management
- Cows
- goats
- management
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Cows and goats management without a clear example, method, or evidence.

Cows and goats management: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Cows and goats management.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Cows and goats management: Practice And Correction

Practice for Cows and goats management:

Main outcome: explain how to care for cows and goats properly and show good habits in keeping animals that help families and communities.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Cows and goats management: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Cows and goats management.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Cows and goats management: Outcome Check

Outcome check for Cows and goats management: Explain how to care for cows and goats properly and show good habits in keeping animals that help families and communities.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Cows and goats management: Outcome Check

Outcome check for Cows and goats management: Name different types of cows and goats and their uses.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Word Processing and Spreadsheet: Lesson Opener

Learning focus: Word Processing and Spreadsheet

Country link: a safe Rwandan observation, model, data table, or investigation for Word Processing and Spreadsheet.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- m anipulate documents using a Word Processing program and perform mathematical calculations using a Spreadsheet program and format a spreadsheet document
- describe how to open, save and close Word Processing and Spreadsheet
- recognize the formatting tools in AbiWord or Word and use them
- open, edit and save documents in AbiWord (or Ms Word) and Spreadsheets
- format text using bold, italics, underline and change font properties for better text readability
- appreciate the usefulness and aesthetics of text and spreadsheet formatting

Inquiry questions:
- How can word processing and spreadsheet help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Word Processing and Spreadsheet: Learn

Word Processing and Spreadsheet teaches useful computer work through exact steps, not guessing.

Core idea:
- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:
- Word Processing and Spreadsheet
- Word
- Processing
- Spreadsheet
- concept
- observation
- evidence
- safety

Word Processing and Spreadsheet: Worked Example

Worked example for Word Processing and Spreadsheet: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Word Processing and Spreadsheet: Vocabulary And Study Skill

Use these words and study moves before you practise Word Processing and Spreadsheet. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Word Processing and Spreadsheet
- Word
- Processing
- Spreadsheet
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Word Processing and Spreadsheet without a clear example, method, or evidence.

Word Processing and Spreadsheet: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Word Processing and Spreadsheet.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Word Processing and Spreadsheet: Practice And Correction

Practice for Word Processing and Spreadsheet:

Main outcome: m anipulate documents using a Word Processing program and perform mathematical calculations using a Spreadsheet program and format a spreadsheet document.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Word Processing and Spreadsheet: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Word Processing and Spreadsheet.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Word Processing and Spreadsheet: Outcome Check

Outcome check for Word Processing and Spreadsheet: M anipulate documents using a Word Processing program and perform mathematical calculations using a Spreadsheet program and format a spreadsheet document.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Word Processing and Spreadsheet: Outcome Check

Outcome check for Word Processing and Spreadsheet: Describe how to open, save and close Word Processing and Spreadsheet.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Lesson Opener

Learning focus: Programming for children

Country link: a safe Rwandan observation, model, data table, or investigation for Programming for children.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- use Scratch to design shapes, do calculations, and create simple animations with sprites and backgrounds
- identify turtle art instruction to draw cylinder, cube, cuboids
- summarize a given story using animations
- construct different geometric shapes using turtle art instruction or Scratch
- perform different calculations in Scratch
- summarize a given story using animations

Inquiry questions:
- How can programming for children help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Learn

Programming for children is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:
- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:
- Programming for children
- Programming
- children
- concept
- observation
- evidence
- safety
- conclusion

Programming for children: Worked Example

Worked example for Programming for children: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Programming for children: Vocabulary And Study Skill

Use these words and study moves before you practise Programming for children. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Programming for children
- Programming
- children
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Programming for children without a clear example, method, or evidence.

Programming for children: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Programming for children.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Programming for children: Practice And Correction

Practice for Programming for children:

Main outcome: use Scratch to design shapes, do calculations, and create simple animations with sprites and backgrounds.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Programming for children: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Programming for children.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Outcome Check

Outcome check for Programming for children: Use Scratch to design shapes, do calculations, and create simple animations with sprites and backgrounds.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Programming for children: Outcome Check

Outcome check for Programming for children: Identify turtle art instruction to draw cylinder, cube, cuboids.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants reproduction: Lesson Opener

Learning focus: Plants reproduction

Country link: a safe Rwandan observation, model, data table, or investigation for Plants reproduction.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- identify and explain how plants reproduce, recognize different types of plant reproduction, and show interest in caring for plants around them
- name the parts of a flower for plant reproduction
- explain how flowering plants reproduce sexually
- describe how plants reproduce asexually
- draw and label the parts of a flower
- observe and classify plants based on whether they reproduce sexually or asexuallyDifferentiate sexual reproduction and asexual reproduction

Inquiry questions:
- How can plants reproduction help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants reproduction: Learn

Plants reproduction teaches the human reproductive system with correct vocabulary, respect, privacy, and health awareness.

Core idea:
- Reproductive organs have specific structures and functions.
- Puberty brings physical and emotional changes that should be discussed respectfully.
- Personal hygiene, safety, consent, and seeking trusted adult or health-worker support are important.
- Diagrams should be labelled scientifically and handled with maturity.

Key words:
- Plants reproduction
- Plants
- reproduction
- concept
- observation
- evidence
- safety
- conclusion

Plants reproduction: Worked Example

Model for Plants reproduction: Label and explain a reproductive-system diagram respectfully.

1. Use the scientific diagram provided by the teacher or book.
2. Label the main structures with correct vocabulary.
3. Write one function for each labelled structure.
4. Add one health or hygiene point.
5. Keep discussion respectful and private.

Conclusion: correct labels and respectful explanation help learners understand body systems safely.

Plants reproduction: Vocabulary And Study Skill

Use these words and study moves before you practise Plants reproduction. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Plants reproduction
- Plants
- reproduction
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Plants reproduction without a clear example, method, or evidence.

Plants reproduction: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Plants reproduction.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Plants reproduction: Practice And Correction

Practice for Plants reproduction:

Main outcome: identify and explain how plants reproduce, recognize different types of plant reproduction, and show interest in caring for plants around them.

1. Label a teacher-approved reproductive-system diagram.
2. Write one function for three labelled parts.
3. List two respectful health, hygiene, privacy, or safety points.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Plants reproduction: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Plants reproduction.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants reproduction: Outcome Check

Outcome check for Plants reproduction: Identify and explain how plants reproduce, recognize different types of plant reproduction, and show interest in caring for plants around them.

1. Label three structures on a teacher-approved diagram.
2. Write one function for each.
3. Add one respectful health or hygiene point.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Plants reproduction: Outcome Check

Outcome check for Plants reproduction: Name the parts of a flower for plant reproduction.

1. Label three structures on a teacher-approved diagram.
2. Write one function for each.
3. Add one respectful health or hygiene point.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Sustainable waste management: Lesson Opener

Learning focus: Sustainable waste management

Country link: a safe Rwandan observation, model, data table, or investigation for Sustainable waste management.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- understand different types of waste, use ways to manage waste safely, and show care for the environment by handling waste properly
- state the difference between biodegradable and non- biodegradable waste
- name different sources of waste (like homes, schools, markets)
- observe and distinguish between biodegradable and non- biodegradable waste in their surroundings. Investigate different ways people manage waste in their community
- be careful about dangerous waste like inflammable, toxic, or corrosive materials
- show responsibility by separating hazardous, organic, and recyclable waste

Inquiry questions:
- How can sustainable waste management help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Sustainable waste management: Learn

Sustainable waste management is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Sustainable waste management
- Sustainable
- waste
- management
- concept
- observation
- evidence
- safety

Sustainable waste management: Worked Example

Investigation model for Sustainable waste management: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Sustainable waste management: Vocabulary And Study Skill

Use these words and study moves before you practise Sustainable waste management. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Sustainable waste management
- Sustainable
- waste
- management
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Sustainable waste management without a clear example, method, or evidence.

Sustainable waste management: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Sustainable waste management.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Sustainable waste management: Practice And Correction

Practice for Sustainable waste management:

Main outcome: understand different types of waste, use ways to manage waste safely, and show care for the environment by handling waste properly.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Sustainable waste management: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Sustainable waste management.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Sustainable waste management: Outcome Check

Outcome check for Sustainable waste management: Understand different types of waste, use ways to manage waste safely, and show care for the environment by handling waste properly.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Sustainable waste management: Outcome Check

Outcome check for Sustainable waste management: State the difference between biodegradable and non- biodegradable waste.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human circulatory system: Lesson Opener

Learning focus: Human circulatory system

Country link: a safe Rwandan observation, model, data table, or investigation for Human circulatory system.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- identify parts of the circulatory system, describe how it works, and develop healthy habits to keep it working well
- investigate and explain human circulatory system using evidence
- explain how to keep the circulatory system healthy
- identify common diseases of the circulatory system
- draw and label the human circulatory system
- practice good hygiene to keep the circulatory system healthy

Inquiry questions:
- How can human circulatory system help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human circulatory system: Learn

Human circulatory system is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Human circulatory system
- Human
- circulatory
- system
- concept
- observation
- evidence
- safety

Human circulatory system: Worked Example

Investigation model for Human circulatory system: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Human circulatory system: Vocabulary And Study Skill

Use these words and study moves before you practise Human circulatory system. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Human circulatory system
- Human
- circulatory
- system
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Human circulatory system without a clear example, method, or evidence.

Human circulatory system: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Human circulatory system.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Human circulatory system: Practice And Correction

Practice for Human circulatory system:

Main outcome: identify parts of the circulatory system, describe how it works, and develop healthy habits to keep it working well.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Human circulatory system: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Human circulatory system.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human circulatory system: Outcome Check

Outcome check for Human circulatory system: Identify parts of the circulatory system, describe how it works, and develop healthy habits to keep it working well.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human circulatory system: Outcome Check

Outcome check for Human circulatory system: Investigate and explain human circulatory system using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human respiratory system: Lesson Opener

Learning focus: Human respiratory system

Country link: a safe Rwandan observation, model, data table, or investigation for Human respiratory system.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- identify parts of the respiratory system, describe how it works, and practice healthy habits to keep it working well
- explain the main functions of the respiratory system
- identify the parts of the human respiratory system
- explain the process of breathing
- look at charts or models of the respiratory system and describe them
- draw and label the parts of the respiratory system

Inquiry questions:
- How can human respiratory system help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human respiratory system: Learn

Human respiratory system is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:
- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:
- Human respiratory system
- Human
- respiratory
- system
- concept
- observation
- evidence
- safety

Human respiratory system: Worked Example

Worked example for Human respiratory system: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Human respiratory system: Vocabulary And Study Skill

Use these words and study moves before you practise Human respiratory system. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Human respiratory system
- Human
- respiratory
- system
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Human respiratory system without a clear example, method, or evidence.

Human respiratory system: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Human respiratory system.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Human respiratory system: Practice And Correction

Practice for Human respiratory system:

Main outcome: identify parts of the respiratory system, describe how it works, and practice healthy habits to keep it working well.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Human respiratory system: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Human respiratory system.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human respiratory system: Outcome Check

Outcome check for Human respiratory system: Identify parts of the respiratory system, describe how it works, and practice healthy habits to keep it working well.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human respiratory system: Outcome Check

Outcome check for Human respiratory system: Explain the main functions of the respiratory system.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human reproductive system: Lesson Opener

Learning focus: Human reproductive system

Country link: a safe Rwandan observation, model, data table, or investigation for Human reproductive system.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- identify the parts and functions of the human reproductive system, explain hygiene practices, puberty changes, and responsible behaviors, and show positive attitudes toward prevention of diseases and respect for people living with STIs and HIV
- explain the function of the human reproductive system
- identify the male and female reproductive organs
- state the function of genital organs
- demonstrate proper hygiene practices for external genital organs
- practice responsible behaviors in situations related to puberty and sexuality

Inquiry questions:
- How can human reproductive system help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human reproductive system: Learn

Human reproductive system teaches the human reproductive system with correct vocabulary, respect, privacy, and health awareness.

Core idea:
- Reproductive organs have specific structures and functions.
- Puberty brings physical and emotional changes that should be discussed respectfully.
- Personal hygiene, safety, consent, and seeking trusted adult or health-worker support are important.
- Diagrams should be labelled scientifically and handled with maturity.

Key words:
- Human reproductive system
- Human
- reproductive
- system
- concept
- observation
- evidence
- safety

Human reproductive system: Worked Example

Model for Human reproductive system: Label and explain a reproductive-system diagram respectfully.

1. Use the scientific diagram provided by the teacher or book.
2. Label the main structures with correct vocabulary.
3. Write one function for each labelled structure.
4. Add one health or hygiene point.
5. Keep discussion respectful and private.

Conclusion: correct labels and respectful explanation help learners understand body systems safely.

Human reproductive system: Vocabulary And Study Skill

Use these words and study moves before you practise Human reproductive system. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Human reproductive system
- Human
- reproductive
- system
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Human reproductive system without a clear example, method, or evidence.

Human reproductive system: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Human reproductive system.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Human reproductive system: Practice And Correction

Practice for Human reproductive system:

Main outcome: identify the parts and functions of the human reproductive system, explain hygiene practices, puberty changes, and responsible behaviors, and show positive attitudes toward prevention of diseases and respect for people living with STIs and HIV.

1. Label a teacher-approved reproductive-system diagram.
2. Write one function for three labelled parts.
3. List two respectful health, hygiene, privacy, or safety points.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Human reproductive system: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Human reproductive system.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human reproductive system: Outcome Check

Outcome check for Human reproductive system: Identify the parts and functions of the human reproductive system, explain hygiene practices, puberty changes, and responsible behaviors, and show positive attitudes toward prevention of diseases and respect for people living with STIs and HIV.

1. Label three structures on a teacher-approved diagram.
2. Write one function for each.
3. Add one respectful health or hygiene point.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Human reproductive system: Outcome Check

Outcome check for Human reproductive system: Explain the function of the human reproductive system.

1. Label three structures on a teacher-approved diagram.
2. Write one function for each.
3. Add one respectful health or hygiene point.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy management: Lesson Opener

Learning focus: Energy management

Country link: a safe Rwandan observation, model, data table, or investigation for Energy management.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- identify different forms and sources of energy, show how energy changes from one form to another, use energy wisely, and care about using clean energy to protect the environment
- define the term energy
- list different forms of energy
- explain how energy changes from one form to another
- explain why energy is important
- choose the best form of energy to use in different situations

Inquiry questions:
- How can energy management help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy management: Learn

Energy management connects work, power, and energy changes.

Core idea:
- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:
- Energy management
- Energy
- management
- concept
- observation
- evidence
- safety
- conclusion

Energy management: Worked Example

Worked example for Energy management: A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Energy management: Vocabulary And Study Skill

Use these words and study moves before you practise Energy management. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Energy management
- Energy
- management
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Energy management without a clear example, method, or evidence.

Energy management: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Energy management.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Energy management: Practice And Correction

Practice for Energy management:

Main outcome: identify different forms and sources of energy, show how energy changes from one form to another, use energy wisely, and care about using clean energy to protect the environment.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Energy management: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Energy management.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy management: Outcome Check

Outcome check for Energy management: Identify different forms and sources of energy, show how energy changes from one form to another, use energy wisely, and care about using clean energy to protect the environment.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Energy management: Outcome Check

Outcome check for Energy management: Define the term energy.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetism: Lesson Opener

Learning focus: Magnetism

Country link: a safe Rwandan observation, model, data table, or investigation for Magnetism.

Progression focus: prepare for upper-primary transition by using evidence, accuracy, and reflection; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- explain how magnets work, identify different types and use of magnets, make a temporary magnet, and show interest in exploring magnets in daily life
- explain the meaning of a magnet
- recognize that a magnet can push or pull objects
- investigate and explain magnetism using evidence
- explain different types of magnets and magnetic fields
- compare and group magnets, non-magnets, and magnetic materials

Inquiry questions:
- How can magnetism help you solve a real problem in Science and Elementary Technology?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetism: Learn

Magnetism explains magnetic and non-magnetic materials.

Core idea:
- Magnets attract some materials, especially iron and steel.
- A magnet has north and south poles.
- Like poles repel and unlike poles attract.
- Magnetic force can act without touching the object.

Key words:
- Magnetism
- concept
- observation
- evidence
- safety
- conclusion

Magnetism: Worked Example

Investigation model for Magnetism: Test materials with a magnet.

1. Collect a nail, coin, plastic ruler, paper clip, and wood piece.
2. Bring a magnet near each object.
3. Record attracted or not attracted.
4. Group the materials as magnetic or non-magnetic.

Conclusion: iron and steel objects are usually attracted by magnets.

Magnetism: Vocabulary And Study Skill

Use these words and study moves before you practise Magnetism. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Magnetism
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Magnetism without a clear example, method, or evidence.

Magnetism: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Magnetism.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Magnetism: Practice And Correction

Practice for Magnetism:

Main outcome: explain how magnets work, identify different types and use of magnets, make a temporary magnet, and show interest in exploring magnets in daily life.

1. Test five safe materials with a magnet.
2. Record attracted/not attracted in a table.
3. State what happens when like poles are brought together.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Magnetism: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Magnetism.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetism: Outcome Check

Outcome check for Magnetism: Explain how magnets work, identify different types and use of magnets, make a temporary magnet, and show interest in exploring magnets in daily life.

1. Name two magnetic materials.
2. State what happens with like poles.
3. Describe a fair magnet test.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Magnetism: Outcome Check

Outcome check for Magnetism: Explain the meaning of a magnet.

1. Name two magnetic materials.
2. State what happens with like poles.
3. Describe a fair magnet test.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Mechanics tools: Review Clinic 1

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Mechanics tools that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Simple machines: Review Clinic 2

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Simple machines that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Air pollution: Review Clinic 3

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Air pollution that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Cows and goats management: Review Clinic 4

Use this review clinic to strengthen Science and Elementary Technology without copying earlier answers.

1. Choose one idea from Cows and goats management that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Science and Elementary Technology.

- Mechanics tools: Explain this term using an example from the book, your school, or your community.
- Simple machines: Explain this term using an example from the book, your school, or your community.
- Air pollution: Explain this term using an example from the book, your school, or your community.
- Cows and goats management: Explain this term using an example from the book, your school, or your community.
- Word Processing and Spreadsheet: Explain this term using an example from the book, your school, or your community.
- Programming for children: Explain this term using an example from the book, your school, or your community.
- Plants reproduction: Explain this term using an example from the book, your school, or your community.
- Sustainable waste management: Explain this term using an example from the book, your school, or your community.
- Human circulatory system: Explain this term using an example from the book, your school, or your community.
- Human respiratory system: Explain this term using an example from the book, your school, or your community.
- Human reproductive system: Explain this term using an example from the book, your school, or your community.
- Energy management: Explain this term using an example from the book, your school, or your community.
- Magnetism: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Science and Elementary Technology answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Science and Elementary Technology that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.